

Course «C Programming Language»

Topic: Multidimensional dynamic arrays

Homework

Exercise 1.

Write a function that adds a column in two-dimensional array to a specified position.

Exercise 2.

Write a function that removes a column of two-dimensional array according to a specified number.

Exercise 3.

Given a matrix of $M \times N$ elements (M rows, N columns). You must fill it with values and write a function that implements a circular shift of rows and / or columns of the array specified number of times in a specified direction.

Exercise 4.

The transpose of given matrix is formed by interchanging rows and columns of matrix. Write a function of transpose of matrix.

Exercise 5.

Create a dynamic array that contains a name in a first string and a phone number in a second string. Organize a search according to the name and phone number, and possibility to enter and modify data.

Course «C Programming Language»

Topic: Structures

Homework

Exercise 1.

Create a structure that describes a complex number. Implement arithmetic operations with complex numbers: addition, subtraction, multiplication, division.

Exercise 2.

Implement a structure "Car" (length, clearance (ride height), engine size, engine power, diameter of wheels, color, gearbox type). Create functions to specify the values, to display the values, to search for the values.

Exercise 3.

Develop a program "Library". Create a structure "Book" (title, author, publisher, genre). Create an array of 10 books. Implement the following options for it:

- edit book;
- print all books;
- search for books by author;
- search for books by title;
- sort an array by title of books;
- sort an array by author;
- sort an array by publishing house.

Exercise 4.

Implement a structure "Car" (color, model, number). Car number can be either a five-digit number or a word up to 8 characters.

Recommendations: implement a number as union.

Create a copy of the structure "Car" and implement the following functions for it:

- fill car;
- print car.

Create an array of 10 copies of the structure "Car" and implement the following functions for it:

- edit car;
- print all cars;
- search for car by number.